As a pathologist, I studied disease through a microscope.

As an epidemiologist, I studied patterns across communities.

Then it became personal.

After I lost my job, my family lost our health insurance. The next month, my husband was bitten by a tick and became ill with Lyme disease.

I understood disease. But as a wife, I faced another question: How do we find our way through this?

That is why I built PathwayAI.

I'm Yuwei Zhang. Let me show you.

For the TOPx HHS Cost of Illness challenge, our goal is to make the hidden burden before diagnosis visible. This working pilot website connects practical help for patients with evidence for community planning.

Reducing burden starts before a bite—with informed choices and prevention. After a bite, people need timely advice and a way into care.

Early treatment can help prevent Lyme disease from spreading and reduce serious complications. Practical support can help people reach that care. Reducing disability and future costs is our goal, not a result we have demonstrated.

What if you don't have insurance?

Here, you can find community health centers serving uninsured patients, ask about income-based fees, and explore coverage or hospital financial assistance.

Care is not automatically free. But you have a starting point, without sharing a story with PathwayAI.

Now imagine someone feeling unwell and preparing for an appointment. This example is fictional.

Write rough notes about symptoms, dates, tests, and questions. PathwayAI helps organize them into a summary you can check, download, and bring to different clinicians.

Optional AI helps organize your words. The basic organizer works without external AI.

A clear record can make it easier to explain what happened across appointments.

Here, the person paid two hundred twenty-five dollars for care and travel. They missed three workdays. A caregiver contributed eight hours.

The burden is more than a medical bill. It includes travel, missed work, and unpaid care.

We keep these separate. Unknown answers stay unknown.

The person can also prepare a support request about transport eligibility, cost, booking, and the return trip.

For example, Dutchess older adults without Medicaid can ask the Office for the Aging about medical-ride registration and availability.

Now we step back—from one household to the community.

Public data for Dutchess and Orange counties show reported Lyme disease, population context, and healthcare resources.

Published cost research illustrates possible burden. Sources and assumptions stay visible. These are planning scenarios, not measured county spending or estimates of the full pre-diagnosis burden.

What could a community do next?

Patients and partners could review transport, affordable care, and navigation barriers, then choose a feasible response.

But identifying a gap does not create funding.

The planning brief helps discuss capacity, staffing, costs, and possible funding sources. It could inform a county budget discussion or grant proposal.

Then ask: Did people reach care? Did assistance come through? Who still needs help?

These invented records show how comparable reports could reveal recurring needs. Shared collection is off. A real pilot needs consent, privacy protections, and accessible participation. Voluntary reports cannot represent everyone.

AI can organize information—but people decide what matters.

Data can reveal patterns—but patients help us understand them.

Our aim is more coordinated, patient-centered care and better-informed community action, with less avoidable disruption and potentially lower costs.

That is PathwayAI: Full Economic Burden.

Making the Hidden Patient Journey Visible.

To patients, caregivers, and experts: thank you. Your experiences and guidance helped make this possible.